Accessible Technology Handbook

A short guide to assistive tools for people with visual impairments

Introduction

Digital accessibility means designing documents, websites and software so that everyone can use them, including people who are blind or have low vision. Screen readers, magnifiers, braille displays and text-to-speech tools turn visual content into sound, enlarged text or touch.

This handbook is a test document for an offline document reader. It contains headings, paragraphs, lists, tables, captions and repeated headers and footers, so that every part of a reading pipeline can be checked.

Why accessibility matters

According to the World Health Organization, at least 2.2 billion people worldwide have a near or distance vision impairment. Many of them rely on assistive technology every day to read mail, study, work and move around safely.

Chapter 1: Screen Readers

A screen reader is software that reads out loud what appears on the screen. It follows the structure of a document, so headings, lists and tables must be marked up correctly for it to work well.

Common screen readers

- NVDA is a free and open-source screen reader for Windows.
- JAWS is a commercial screen reader for Windows.
- VoiceOver is built into macOS and iOS devices.
- TalkBack is built into Android devices.

Navigation shortcuts

Users move through a document by headings, links and landmarks rather than line by line. Pressing the H key in NVDA jumps to the next heading, which can save several minutes in a long report.

> *“A document that cannot be navigated by structure is, for a blind reader, a wall of sound.”*

Chapter 2: Optical Character Recognition

Optical character recognition, or OCR, converts images of text into machine-readable text. It is essential for scanned books and photographed letters, which contain no real text for a screen reader to speak.

How OCR works

1 The page image is converted to grayscale.

2 The image is straightened and thresholded to separate text from background.

3 Lines and words are detected and segmented.

4 Each character is recognised and the text is assembled in reading order.

Comparison of common tools

Tool	Type	Cost	Works offline
NVDA	Screen reader	Free	Yes
JAWS	Screen reader	Paid	Yes
Tesseract	OCR engine	Free	Yes
Cloud OCR service	OCR engine	Paid	No

Table 1: Assistive tools compared by type, cost and offline support.

Tesseract was first developed in the 1980s and is maintained as an open-source project. Accuracy depends strongly on scan quality: a clean 300 DPI scan typically gives far better results than a blurry photo.